

High-Density Pulsed Electromagnetic Power Generator (HDEP-PG)

Technical Design Specifications (v2)

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1 System Architecture

The system utilizes a central pulse-driven magnetic core (20cm) surrounded by a 49-coil independent pick-up array. The electromagnetic architecture integrates inductive coupling with thermoelectric recovery.

2 Input/Output Matrix

The following table defines the operational parameters of the system, including updated target outputs.

Table 1: Operational Parameters & Targets

Parameter	Value	Unit
Input Voltage (V_{in})	24	Volts (DC)
Input Peak Current (I_{in})	20	Amperes
Switching Frequency (f)	1	kHz
Magnetic Core Length	20	cm
Series Array (29 Coils) Target	High-Voltage	$V_{out_HV} > 500$ V
Parallel Array (20 Coils) Target	High-Current	$I_{out_HC} > 50$ A
Secondary Energy Recovery	Thermo-Electric	P_{TEG}

3 Governing Physics

3.1 Inductive Kickback (Back-EMF)

The core mechanism is defined by Faraday's Law of Induction, where the sudden current break induces a high-voltage spike:

$$\varepsilon = -L \frac{di}{dt} \quad (1)$$

Where:

- ε is the induced electromotive force.
- L is the self-inductance of the primary coil.
- $\frac{di}{dt}$ represents the rapid decay of current during the switching break.

3.2 Thermal Energy Harvesting

Waste thermal energy from the *SiC* casing is converted via the Seebeck Effect:

$$P_{TEG} = \alpha \cdot \Delta T \cdot I_{TEG} \quad (2)$$

Where α is the Seebeck coefficient and ΔT is the temperature gradient maintained by the ceramic insulation.

4 System Constraints & Materials

- **Core Material:** Laminated iron. (*Note: Thickness adjusted to 0.15mm – 0.35mm from the original 3mm to properly mitigate eddy currents at 1kHz*).
- **Dielectric Housing:** Silicon Nitride (Si_3N_4) for thermal stability and high-voltage insulation.
- **Thermal Interface:** Silicon Carbide (*SiC*) coupled with thermoelectric modules for hybrid energy harvesting.

5 Engineering & Technical Considerations

The following technical notes have been added to optimize the system for practical implementation and manufacturing:

5.1 Core Lamination Optimization

While the initial specification proposed a 3mm laminated iron core, this thickness is excessive for a 1kHz switching frequency and would result in significant core losses and overheating. It is strongly recommended to utilize **electrical silicon steel with a thickness between 0.15mm and 0.35mm**. This adjustment is critical to effectively suppress eddy currents.

5.2 Switching and Protection Circuitry

Given the rapid interruption of a 20A current to maximize the $\frac{di}{dt}$ factor, the switching components will face immense electrical stress from the back-EMF.

- **Solid-State Switches:** Employ high-speed, high-voltage tolerance components such as **SiC MOSFETs** or **IGBTs** to handle the abrupt transitions.
- **Transient Protection:** The integration of **Snubber Circuits** across the switches is mandatory to suppress destructive voltage spikes and prevent component failure.

5.3 Coil Array Wire Specifications

The 49-coil array requires differentiated wire gauges (AWG) and insulation ratings based on their specific loads:

- **Series Array (High-Voltage):** Requires wire with advanced, high-dielectric-strength insulation to prevent internal arcing and insulation breakdown under extreme potential differences.
- **Parallel Array (High-Current):** Requires a thicker wire gauge (lower AWG number) to safely conduct high amperage without excessive resistive heating (I^2R losses) or melting.

5.4 Thermal Gradient Definition (ΔT)

For accurate estimation of thermoelectric power (P_{TEG}), the temperature gradient (ΔT) must be defined in the thermal model. Future iterations of this blueprint should specify the expected steady-state operating temperature of the SiC casing (T_{hot}) relative to the ambient or active heat sink temperature (T_{cold}).